[redacted] Conveyor Defect Detection

[redacted]: «[redacted]»
GitHub: https://github.com/rvaleev2016-eng/conveyor-defect-detection
ClearML Showcase: https://app.clear.ml/projects/1b01d7ca2ca741fb881df31ee748a1e6/experiments/0db5742cb68f4c46919dcb64fed478e7/output/log

1. [redacted]

[redacted]. [redacted], [redacted].

2. [redacted] U-Net

U-Net [redacted]: [redacted], [redacted].

3. [redacted]

[redacted]	[redacted]	Dice	IoU
bottle	[redacted]	0.606644	0.521746
capsule	[redacted]	0.178571	0.178571
metal_nut	[redacted]	0.287587	0.228832
pill	[redacted]	0.466299	0.359749
bottle	[redacted]	0.1875	0.1875
capsule	[redacted]	0.178571	0.178571
metal_nut	[redacted]	0.125	0.125
pill	[redacted]	0.166667	0.166667

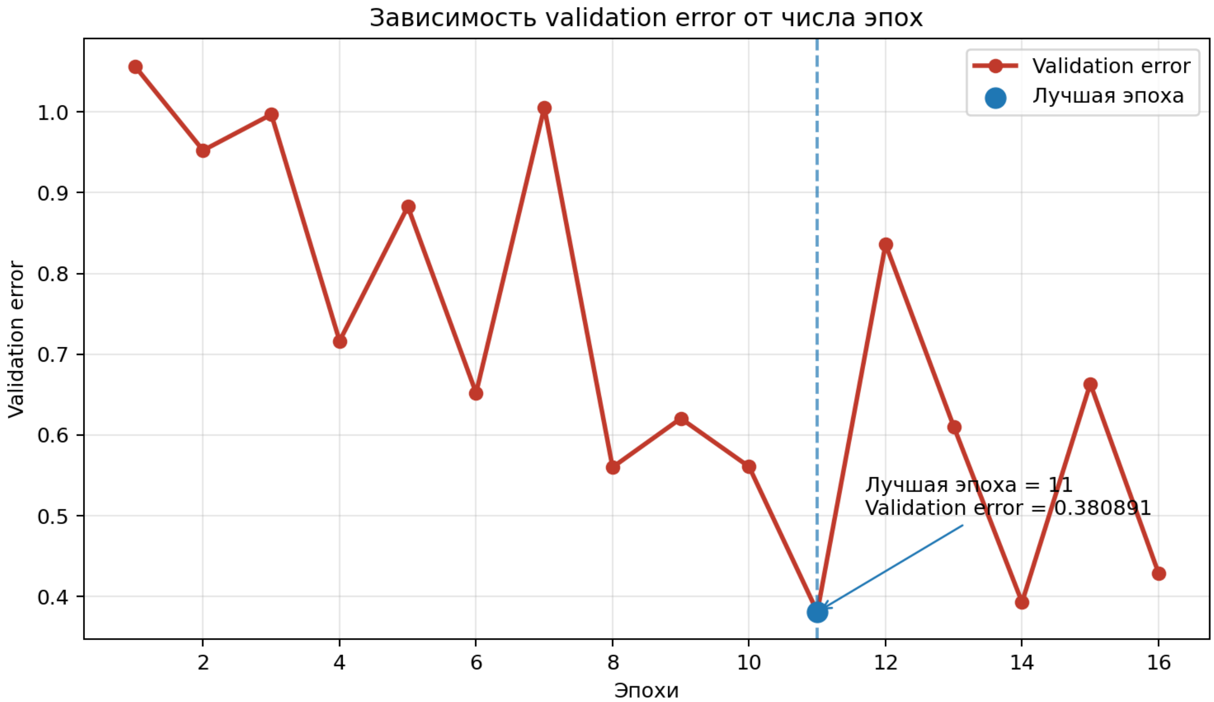
[redacted]: [redacted] bottle [redacted] Dice [redacted] IoU.

4. Bottle: [redacted]

[redacted]	[redacted]
Dice	0.538667
IoU	0.42929
Threshold	0.5

5. Validation error [redacted]

[redacted] validation error. [redacted] learning rate [redacted].

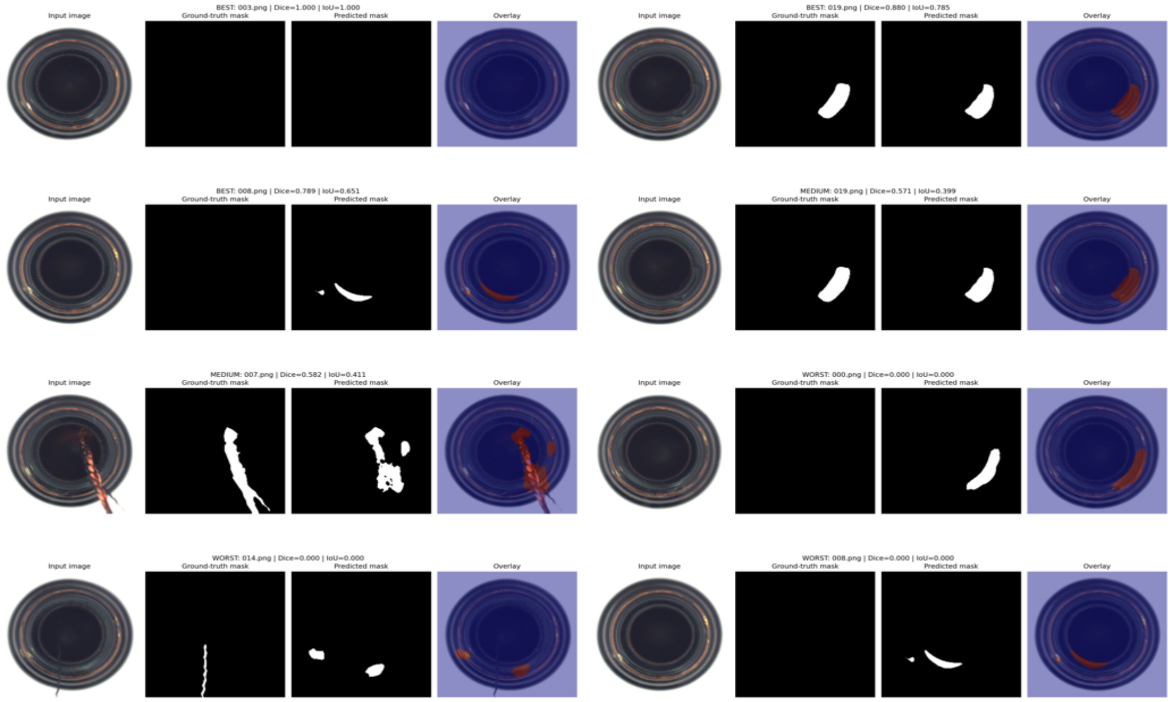


[redacted]	[redacted]	[redacted]	[redacted] validation error	[redacted]
bottle	per-category	10	0.56177	12
capsule	per-category	12	0.798049	12
metal_nut	per-category	11	0.35059	12
pill	per-category	9	0.643102	12
all_in_one	all-in-one	4	0.874632	9

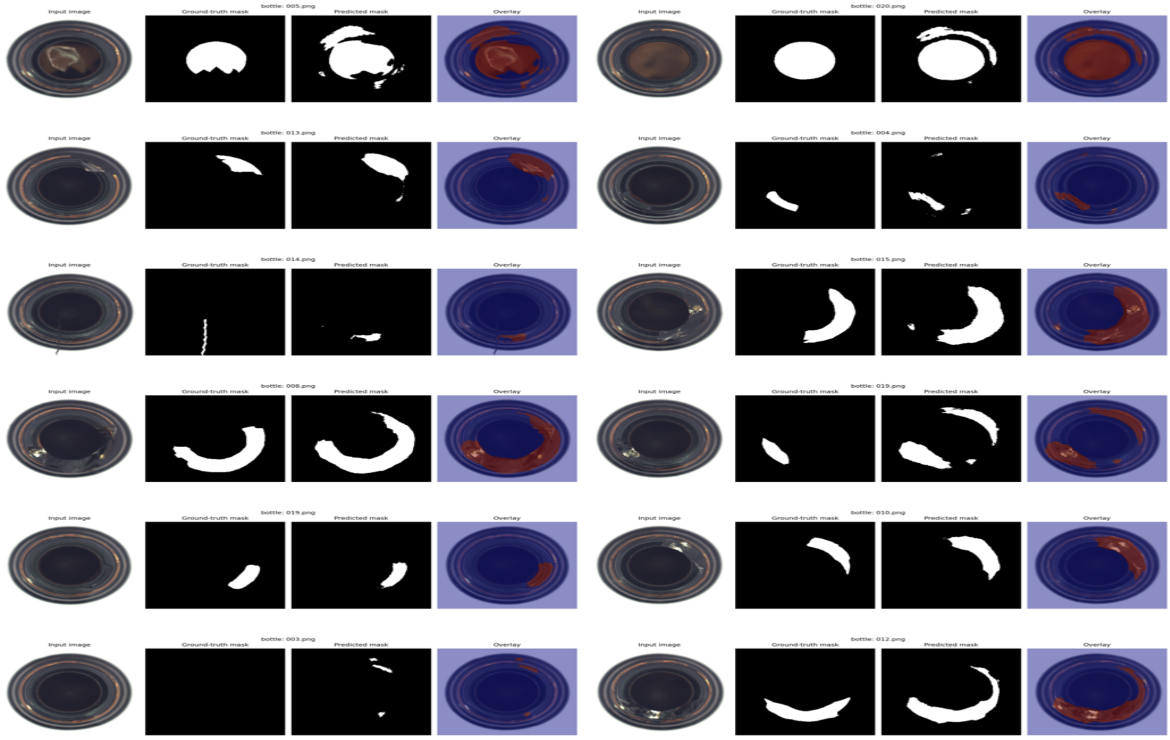
6. [redacted]

[redacted], [redacted].

[redacted], [redacted] bottle

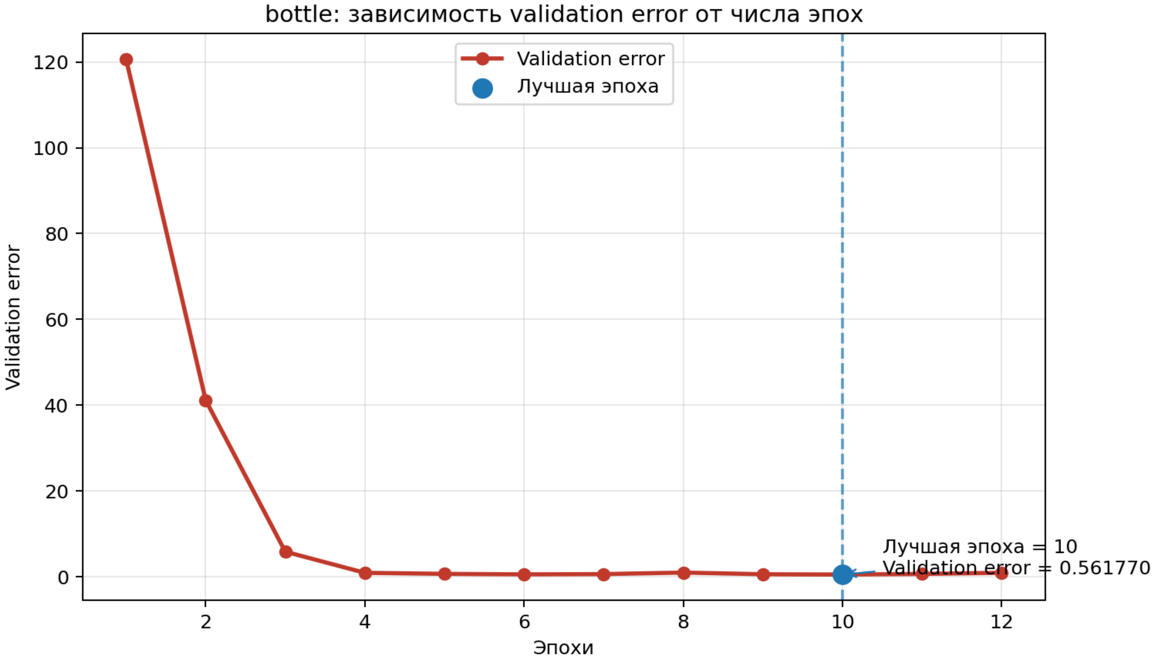


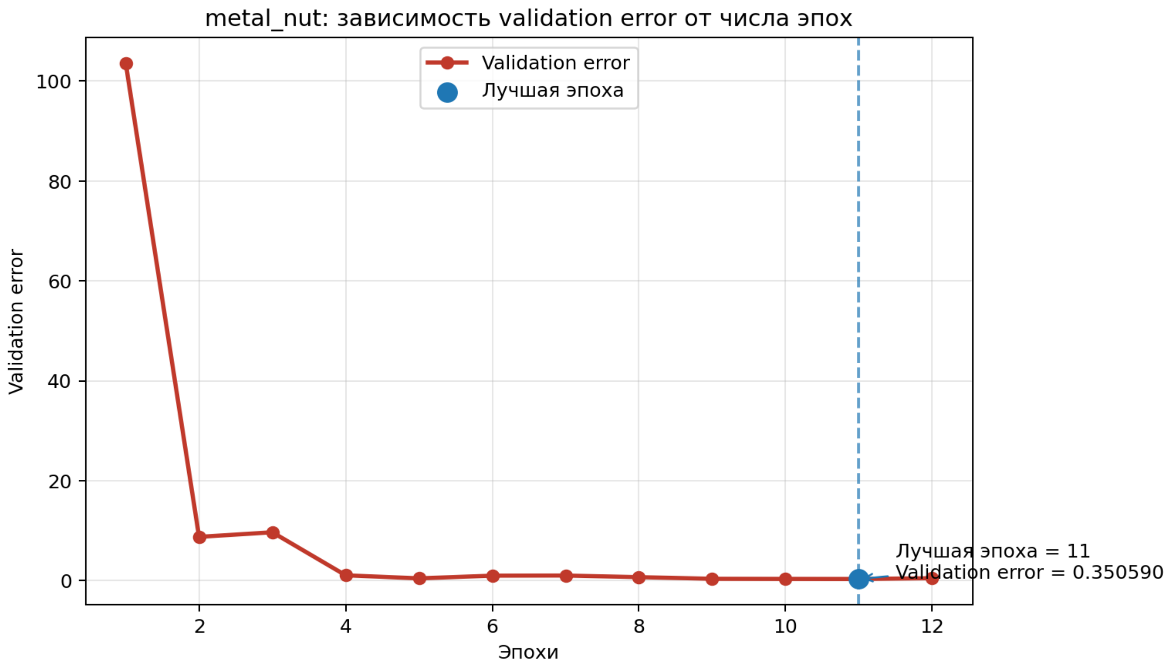
[illegible] bottle

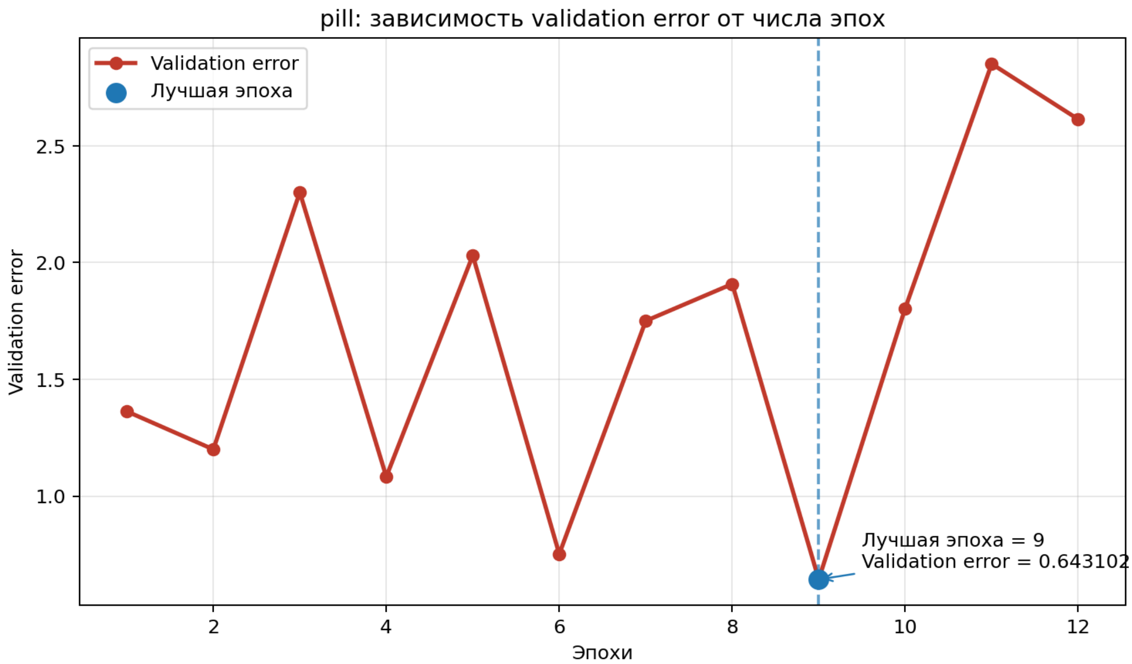


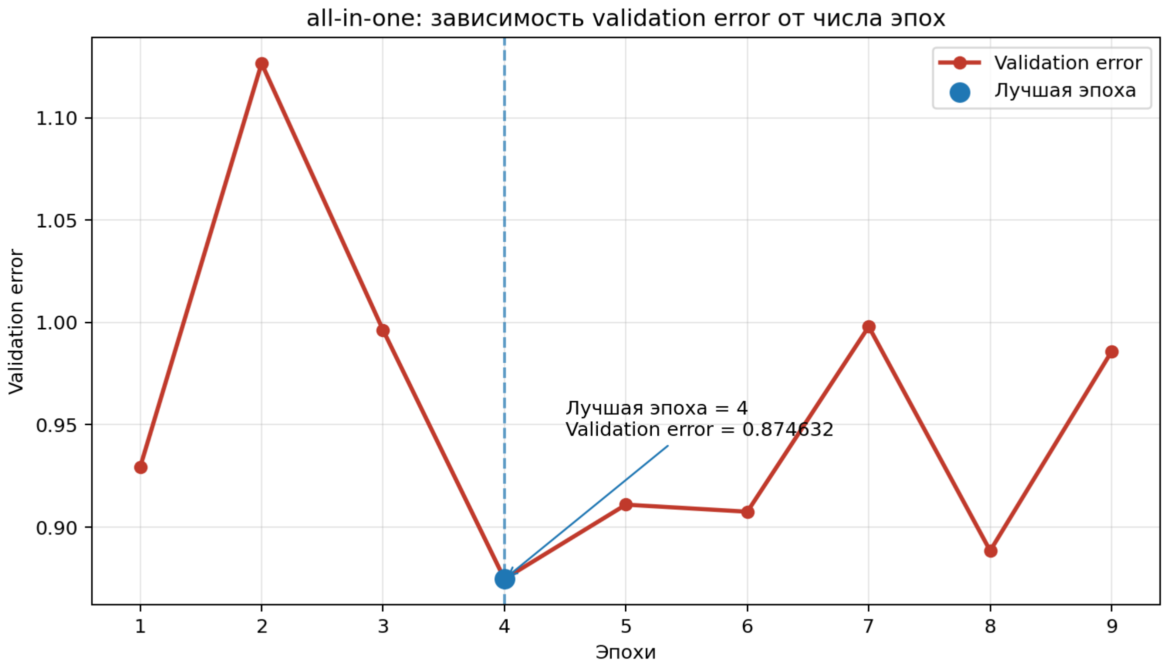
Input image
Ground-truth mask
Predicted mask
Overlay
bottle: 005.png
bottle: 020.png
bottle: 013.png
bottle: 004.png
bottle: 014.png
bottle: 015.png
bottle: 008.png
bottle: 019.png
bottle: 019.png
bottle: 010.png
bottle: 003.png
bottle: 012.png

7. [redacted] validation error [redacted]




metal_nut: зависимость validation error от числа эпох
Validation error
Лучшая эпоха
Лучшая эпоха = 11
Validation error = 0.350590
Validation error
Эпохи
0
20
40
60
80
100
2
4
6
8
10
12


pill: зависимость validation error от числа эпох
Validation error
Лучшая эпоха
Лучшая эпоха = 9
Validation error = 0.643102
Validation error
Эпохи
1.0
1.5
2.0
2.5
2
4
6
8
10
12


all-in-one: зависимость validation error от числа эпох
Validation error
Лучшая эпоха
Лучшая эпоха = 4
Validation error = 0.874632
1.10
1.05
1.00
0.95
0.90
Validation error
1
2
3
4
5
6
7
8
9
Эпохи

8. [redacted]

[redacted]: [redacted], README [redacted] /docs, [redacted] ClearML showcase-[redacted]. [redacted], [redacted] GitHub.